IT Business Analysis

Requirements Analysis

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Learning Objectives

- Requirements Analysis

Learning Outcomes

- Learning outcome 4: Professional standard
- Learning outcome 5: Personal Leadership

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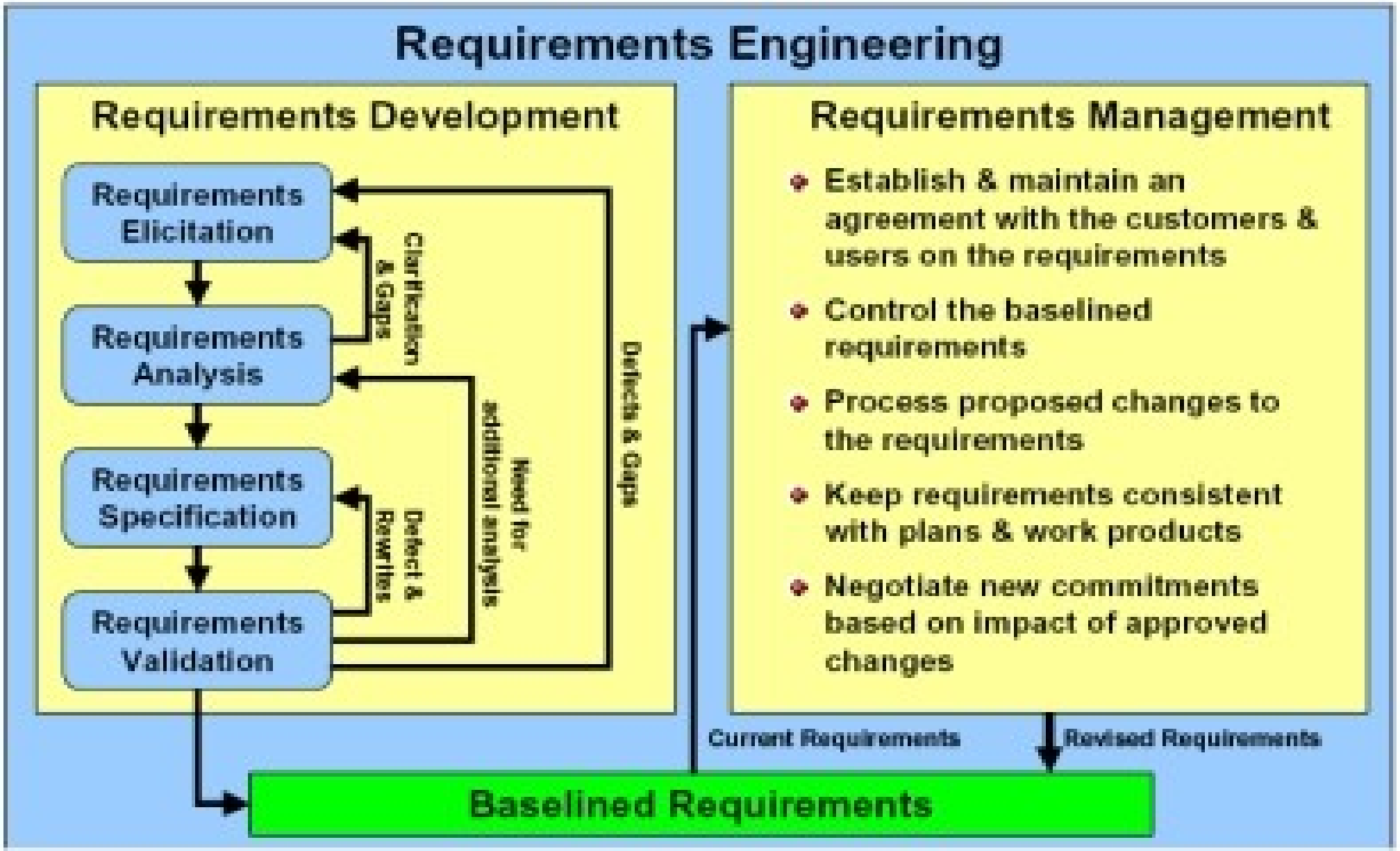
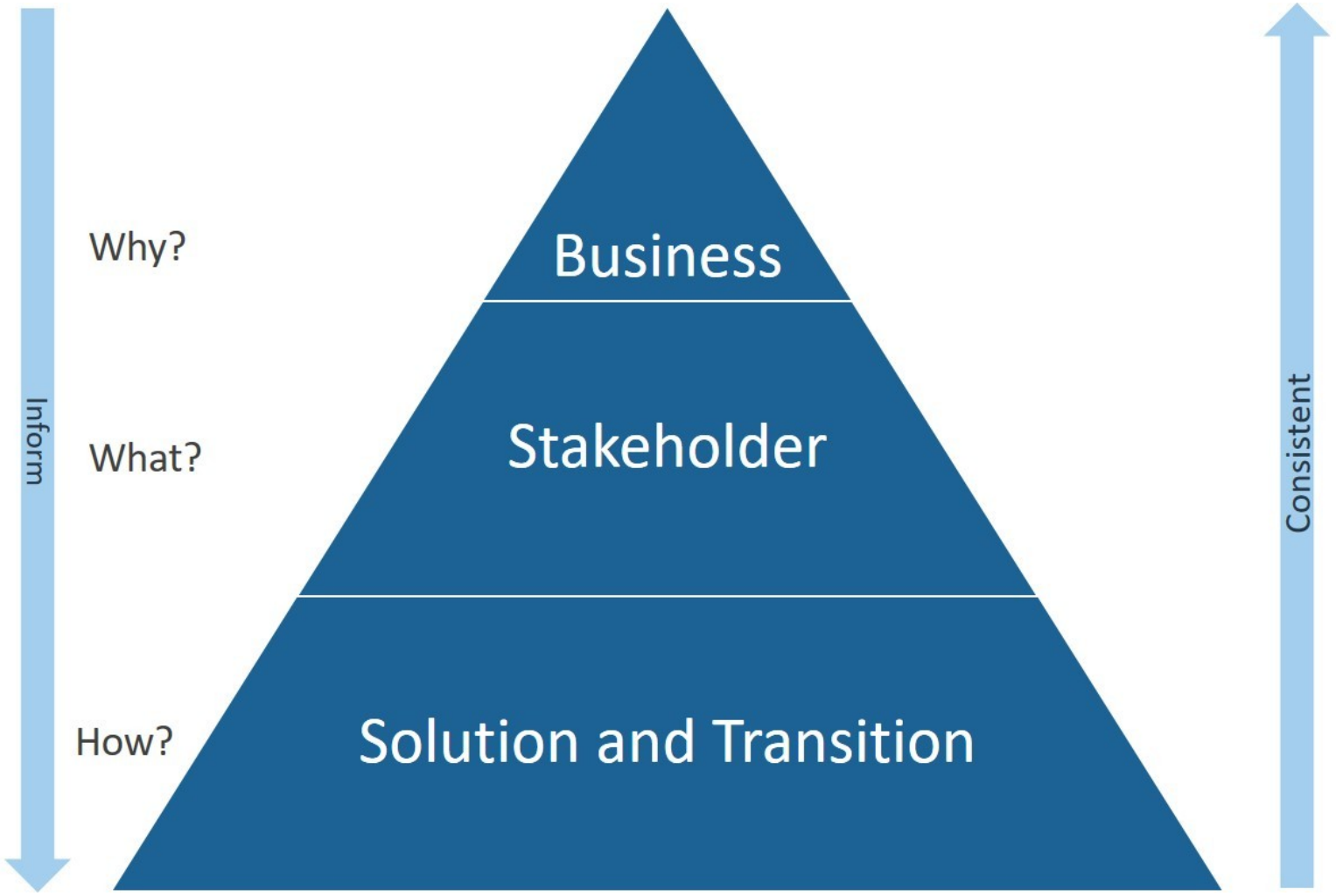


http://www.raamstijn.nl/eenblogjeom/index.php/requirements/198-requirements-engineering

Level of Requirements



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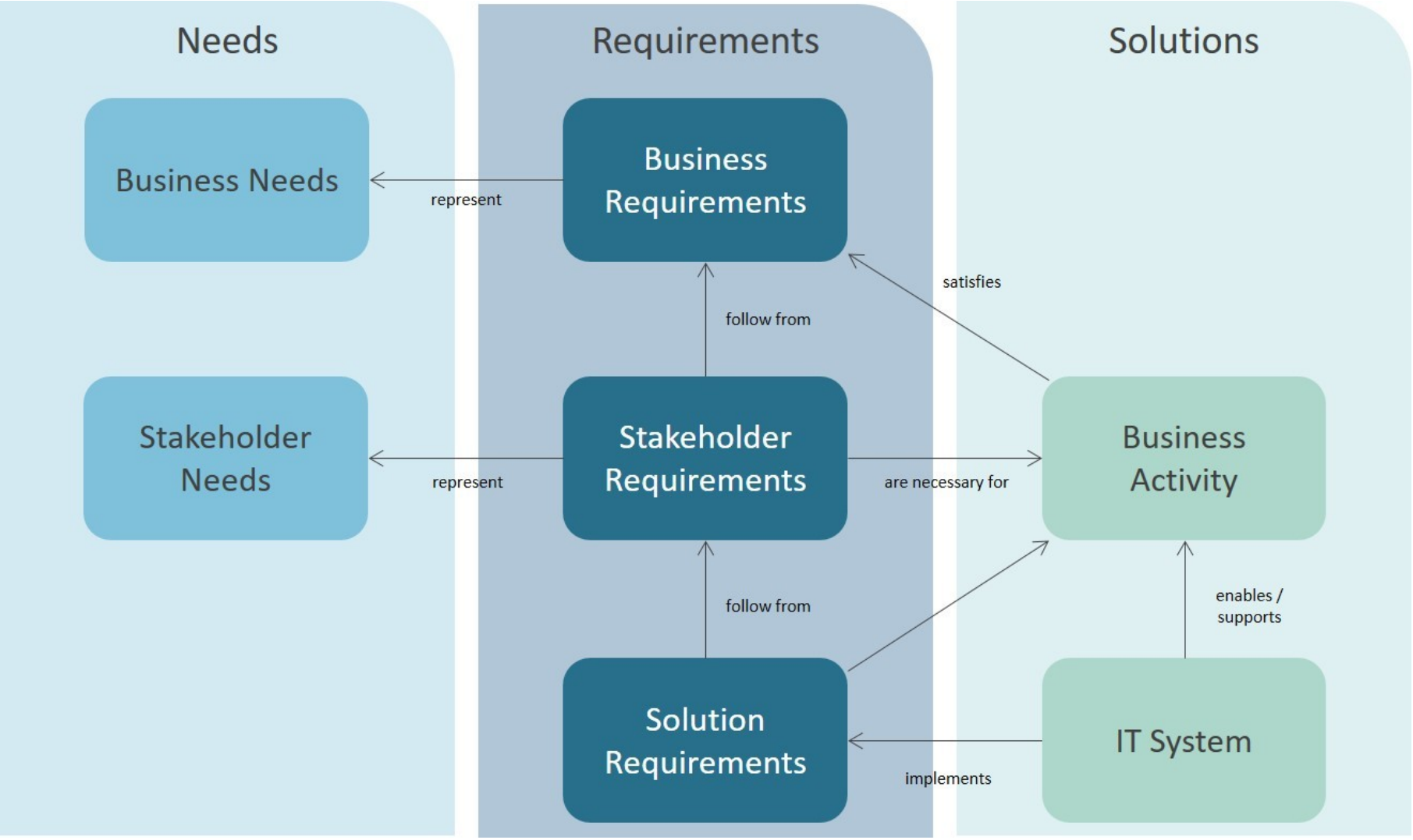
Need vs Requirements



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Business Requirements

Unique ID!!
For each requirement

- Business Requirement → “WHY”
  - Why the organization undertakes the project, needs the system?
  - Comes from Business Analysis, Business Case or Project vision
  - The PROBLEM to solve must be reflected (research question?)

B01 The CFO wants to have 90% of the purchase invoices paid automatically to reduce the workload for the AP department
B02 The CFO wants to have up-to-date visualized financial information in order to be able to evaluate the financial obligations of the AP department

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User Requirements or Stakeholders Requirements

Unique ID!!
For each requirement

- User Requirements → "WHAT"
  - What the user will be able to do with the system (goals/tasks)
  - One functionality per requirement
  - Related to the Business Requirements
  - Difficult, because the user self does not always know what the system/software should do (an important role for Business Analysts)

UR01	The Accounts Payable clerk will be able to see the automatically paid purchase invoices
UR02	The AP clerk will be able to authorize payment for purchase invoices that are not processed automatically
UR03	The Financial Director will be able to slice in months and years on the dashboard graphics

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System Requirements (or Solution Requirements)

Unique ID!!
For each requirement

- System (Functional/Non-function) Requirement → "HOW"
  - Describes what the system shall do (must/should/may), and
  - *what the developer should build?*
  - One functionality per requirement
  - Non-functional requirements must be measurable (SMART)
  - Related to the Business Requirements

SR01	The system shall compare the ordered number of products on the purchase order, with the delivered quantities and the invoiced number of products on the related invoice
SR02	The system shall flag purchase invoices as 'ready for payment' after authorization
SR03	The system shall only pay purchase invoice when they have a purchase order number
SR04	The system shall generate a visual at least within 5 seconds

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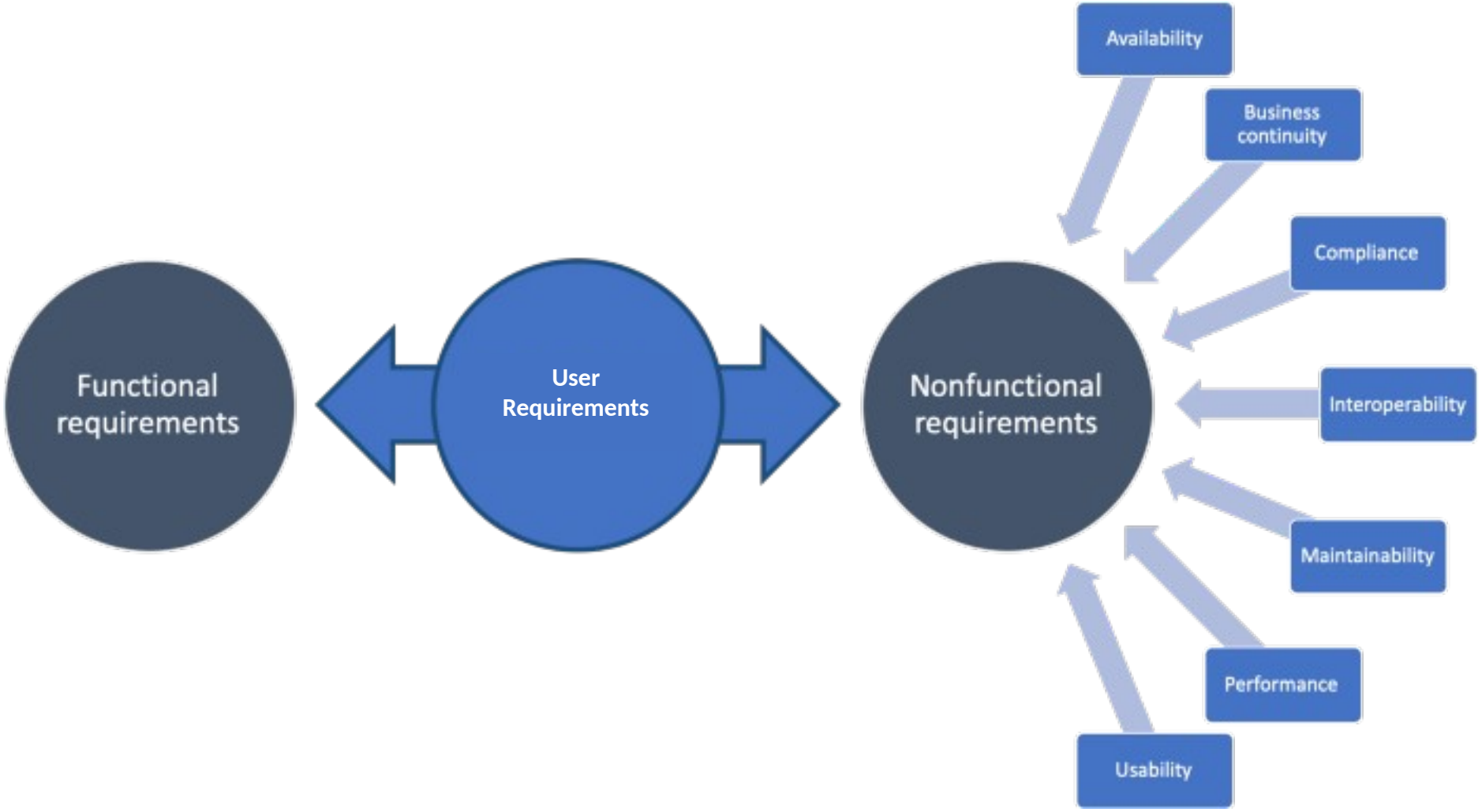
Functional vs Non Functional



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Functional vs Non-Functional

	Functional Requirement	Non-Functional Requirement
Requirement	It is mandatory	It is non-mandatory
Capturing type	It is captured in use case	It is captured as a quality attribute
End-result	Product feature	Product properties
Capturing	Easy to capture	Hard to capture
Objective	Helps you verify the functionality of the software	Helps you to verify the performance of the software
Area of focus	Focus on user requirement	Concentrates on the user's expectation.
Documentation	Describe what the product does	Describes how the product works
Type of Testing	Functional Testing like System, Integration, End to End, API testing, etc.	Non-Functional Testing like Performance, Stress, Usability, Security testing, etc.
Test Execution	Test Execution is done before non-functional testing	After the functional testing
Product Info	Product Feature	Product Properties

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Requirement Analysis

Quality of Requirements

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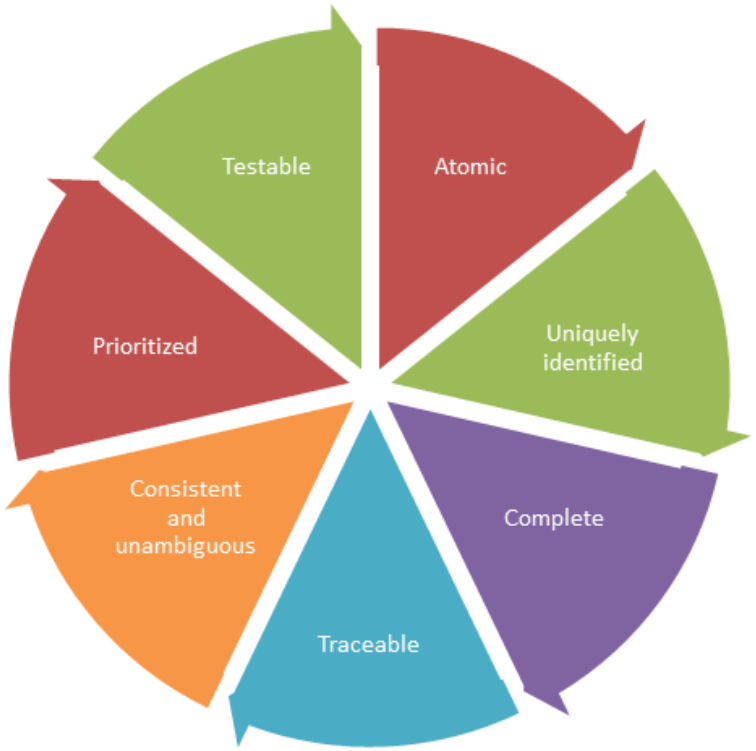
Quality of Requirements

- 7 quality categories
- Using an educational software system where student can register for different courses, as example



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Atomic

Bad Requirement
Requirement

Students will be able to enroll in undergraduate and postgraduate courses

Good

Students will be able to enroll to undergraduate courses

Students will be able to enroll to post-graduate course

Each requirement should address one specific functionality or behavior. Avoid combining multiple functionalities into one requirement.

The bad requirement is not atomic because it talks about two different entities undergraduate and post-graduate courses.

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Uniquely identified and Traceable

Bad Requirement

1- Students will be able to enroll to undergraduate courses

1- Students will be able to enroll to post-graduate courses

Good Requirement

UR1 Students will be able to enroll to undergraduate courses

UR2 Students will be able to enroll to post-graduate courses

Provide unique ID's for your requirements, so that you can refer to them easily. So, we should be able to take every business requirements and map it to the corresponding one or more user- and system requirements!

For the bad requirements, you would not know which requirement is meant when referring to ID 1

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Complete

Bad Requirement

Requirement

A professor user will log into the system by providing his username, password, and other relevant information

Good

A professor user will log into the system by providing his username, password and department code

Each and every requirement should be complete.

The bad requirement says, "and other relevant information".
Here the other relevant information is not clear

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Consistent and unambiguous

Bad Requirement

A student will have either undergraduate courses or post-graduate courses but not both.

Some courses will be open to both under-graduate and post-graduate

Good Requirement

A student will have either undergraduate or post-graduate but not both

For each requirement, you should be consistent and not contradict with other requirements

The problem here is that the first requirement conflicts with the second requirement and it tells that some courses will open to both post-graduate and under-graduate.

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Prioritised

Bad Requirement

Registered student – Priority 1

Maintain User Info – Priority 1

Enroll courses – Priority 1

View Report Card – Priority 1

Good

Register Student – Must have

Maintain User Info – Should have

Enroll courses – Must have

View Report Card – Could have

Every requirement must be prioritized, so the team has guideline which requirement it is able to implement first and which can be done later.

The bad requirement is not prioritized, as every has the same priority

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Testable

Bad Requirement

Each page of the system will load in an acceptable time-frame

Good Requirement

Register student and enroll courses pages of the system will load within 5 seconds

Each requirement should be testable and therefore defined SMART

The bad requirement is "each page....in an acceptable time frame".
Each page means that there can be many pages, which going to blow up the testing efforts. Also what is an acceptable time frame?
Acceptable to whom?

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Recap

- Requirements analysis

Learning Outcomes

- Learning outcome 4: Professional standard
- Learning outcome 5: Personal Leadership

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Thank you!
By Raja Gorentla

Any Question?

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